Tarun Kumar

Portfolio

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Profile

Results-driven **Data Science Analyst** with hands-on expertise in designing and deploying AI-driven solutions across Face Recognition, Object Detection, Natural Language Processing (NLP), and Large Language Models (LLMs). Strong foundation in statistical modelling and time-series forecasting, enabling the development of robust predictive systems and data-driven strategies. Experienced in building scalable, interpretable machine learning pipelines that translate complex data into actionable business insights. Proven ability to align AI initiatives with strategic objectives to drive measurable outcomes.

Skills

AI & Machine Learning: *Machine Learning, Deep Learning, Neural Networks, Model Evaluation, Feature Engineering, Hyperparameter Tuning, TensorFlow, PyTorch, scikit-learn, Keras, Model Interpretability (SHAP, LIME)*

Generative AI & NLP: *Large Language Models (LLMs), Prompt Engineering, Retrieval-Augmented Generation (RAG), OpenAI / Azure OpenAI APIs, Embeddings, Semantic Search, NLP, Named Entity Recognition (NER), Text Classification, Transformers*

Computer Vision: *OpenCV, Object Detection (YOLO), Face Recognition, Image Classification, OCR, Feature Extraction, Image Processing*

Data Science & Analytics: *pandas, NumPy, Statistical Modeling, Time Series Forecasting, Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn, Power BI, Tableau)*

Languages & Databases: *Python, JavaScript, Java, C, C++, SQL, MongoDB, MySQL, PostgreSQL*

Others: *REST APIs (FastAPI / Flask), Docker, Model Deployment, Git, Version Control, MLOps Fundamentals*

Professional Experience

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Data Sciences Analyst (*August 2025 – Present*)

- Developed and deployed computer vision solutions for image processing, OCR, face recognition, and object detection using deep learning frameworks.
- Built intelligent information gathering systems for educational institutions including automated footfall counting, attendance tracking, student recognition, and real-time fight detection with alert mechanisms.
- Designed and implemented retail analytics solutions featuring theft recognition, customer behavior analysis, and employee behavior monitoring to enhance store security and operational efficiency.
- Created time-series forecasting models for restaurant and food court operations to predict daily item sales, optimizing inventory management and reducing food waste.
- Developed Medical Named Entity Recognition (NER) systems to automatically extract and identify diseases, medicines, and medical codes from patient data images and PDFs, improving healthcare data processing efficiency.
- Worked extensively with image processing pipelines, OCR technologies, and advanced ML models for real-world deployment in production environments.

CT University

Assistant Professor (*Aug 2023 –July 2025*)

Handled Responsibilities of Software development for the organization

- I taught undergraduate courses in Web Development, Data Structures, and Software Engineering.
- Guided 50+ students in academic projects focused on MERN stack, system design, and automation.
- Developed internal ERP tools to support academic administration (student registration, result management).
- Designed and delivered lab sessions, assessments, and interactive assignments to improve conceptual clarity.
- Mentored students for university-level project competitions and internal internships.
- Played an instrumental role in automation of exam paper generation using AI-driven tools.

Projects

Medical Named Entity Recognition (NER) System *(NLP, TensorFlow, PyTorch, OCR, LLM)*

- Developed a domain-specific Medical NER system to automatically extract diseases, medications, and medical codes from unstructured patient records.
- Designed OCR pipelines to process scanned prescriptions, PDFs, and medical reports with varying image quality.
- Implemented deep learning–based sequence models (BiLSTM/Transformer architectures) for high-accuracy entity recognition.
- Built robust preprocessing workflows for text normalization, noise removal, and document standardization.
  Enabled healthcare providers to digitize and structure clinical data, significantly reducing manual entry effort and improving retrieval efficiency.

Retail Store Analytics & Security System (Computer Vision, Object Detection, Face Recognition, OpenCV, YOLO, TensorFlow)

- Developed an intelligent computer vision–based surveillance and analytics system for retail environments..
- implemented real-time object detection and tracking models using YOLO for theft detection and suspicious activity monitoring.
- Built face recognition modules for employee monitoring and access control compliance.
- Designed customer behaviour analytics to analyse footfall, shopping patterns, and in-store movement.
- Improved operational security and provided actionable insights for layout optimization and loss prevention strategies.

Restaurant Sales Forecasting System (*(Time Series Forecasting, Machine Learning, Data Analysis, Statistical Modelling)*

- Developed predictive models to forecast daily item-level sales across restaurants and food courts.
- Applied time-series techniques (trend, seasonality, lag features) to identify demand patterns and cyclic behaviour.
- Built automated data pipelines to process historical transactional data from multiple sources.
- Reduced inventory wastage and improved procurement planning through accurate demand forecasting.
- Created interactive visualization dashboards to support management in data-driven decision-making.

Certification

- **Deep Learning (NPTEL, IIT Ropar)** Issued: May 2024 - Link
- **Data Analytics Essentials (Coursera)** Issued: Dec 2023 - Link
- **Calculus for Machine Learning and Data Science (Coursera)** Issued: Nov 2023 – Link
- **Mathematics for Machine Learning and Data Science Specialization (Coursera)** Issued: Nov 2023 – Link
- **Probability & Statistics for Machine Learning & Data Science (Coursera)** Issued: Nov 2023 – Link
- **Linear Algebra for Machine Learning and Data Science (Coursera)** Issued: Sep 2023 – Link

Education

Lovely Professional University – Ludhiana, Punjab *Sept 2021 – July 2023*
M.Tech Computer Science; 9.08
NIT, Agartala – Agartala, Tripura *Aug 2016 – June 2020*
B.Tech ECE; 8.01